

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 1 October 2026, 8:55 AM**State** Finished**Completed on** Thursday, 1 October 2026, 8:57 AM**Time taken** 1 min 15 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int has_pair_with_difference(int arr[], int n, int k) {
5      int i = 0;
6      int j = 1;
7
8      while (i < n && j < n) {
9          if (i == j) {
10             j++;
11             continue;
12         }
13
14         int diff = arr[j] - arr[i];
15
16         if (diff == k) {
17             return 1;
18         } else if (diff < k) {
19             j++;
20         } else {
21             i++;
22         }
23     }
24
25     return 0;
26 }
27
28 int main() {
29     int n, k;
30     if (scanf("%d", &n) != 1) return 0;
31     int *arr = (int *)malloc(n * sizeof(int));
32     if (arr == NULL) return 1;
33     for (int i = 0; i < n; i++) {
34         if (scanf("%d", &arr[i]) != 1) {
35             free(arr);
36             return 1;
37         }
38     }
39     if (scanf("%d", &k) != 1) {

```

```
40         tree(arr);
41         return 1;
42     }
43     printf("%d\n", has_pair_with_difference(arr, n, k));
44     free(arr);
45     return 0;
46 }
47
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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